

URITECT

Bayesian UTI Scoring System

Version 1.1

Model version	uti_bayesian_lr_v1_1_20260926
Target condition	Acute uncomplicated lower urinary tract infection
Fixed prior	50%
Purpose	Technical specification and two-medtech parameter content review
Status	Provisional, literature-derived, and not locally calibrated

Prepared for parameter review by two registered medical technologists. A physician and statistician should additionally review the clinical scope, safety wording, dependencies, and calibration plan.

This document is a thesis research specification. Content validation does not establish diagnostic accuracy or calibration against urine culture.

Implemented model: `uti_bayesian_lr_v1_1_20260926` Method: Provisional grouped likelihood-ratio Bayesian research estimate Prior: 50% for the narrowly gated population only Status: Frozen app implementation; literature-derived and not clinically calibrated

Plain explanation: URIRECT can calculate a research estimate only for a narrowly defined group of symptomatic women. For an eligible calculation, the app shows the provisional percentage, the starting prior, and the exact evidence factors used. It is not a diagnosis, treatment recommendation, or Low/Moderate/High UTI category.

1. What Changed From Version 1.0

Review finding	Version 1.1 correction
Combined model was not validated as one probability model	Calls it provisional and literature-derived; explicitly states the independence assumption and validation requirement.
Visible hematuria could duplicate dipstick blood	Removed its LR from the equation; retained it as consultation context.
Vaginal findings came from a different population	Removed their LRs from the equation; either finding stops ordinary calculation and routes to consultation.
The 50% prior was too broadly applicable	Added six required population confirmations and an acute urinary-symptom gate.
Analyte positivity was too broad	Added exact URIRECT-to-study threshold mapping.
20% and 80% bands were invented	Removed every probability band and UTI likelihood label.
Unknown could be interpreted as absent	Defined unchecked, unknown, unavailable, and unreliable as no update.
Nausea alone triggered an urgent pathway	Prompt action now requires an approved combination or serious safety condition.

2. Intended Population Gate

The estimate is calculated only when all six confirmations and at least one acute urinary symptom are present.

Gate ID	Required confirmation
E1	Patient is a woman.
E2	Patient is 18 to 64 years old.
E3	Patient is confirmed not pregnant.
E4	Patient does not have a urinary catheter.
E5	Patient has no known urinary tract abnormality.
E6	Patient is not immunocompromised.
E7	At least one acute urinary symptom is reported: dysuria, urgency, frequency, suprapubic pain, or visible hematuria.

The ordinary calculation is stopped when vaginal discharge, vaginal irritation, fever/chills, flank pain, or nausea/vomiting is reported. The app then shows a consultation action instead of an ordinary lower-UTI estimate.

If a required confirmation is unchecked, its state is not confirmed or unknown. It is not assumed to be false. The app displays:

This UTI screening estimate is not designed for your reported situation. Consultation with a healthcare professional is suggested.

3. Exact Dipstick Mapping

The frozen ten-analyte k-NN model uses the labels below. Source-study positivity follows Little et al.: leukocyte esterase + or greater and blood haemolysed trace or greater.

Analyte	URIRECT label	Treatment in Group A
Nitrite	Positive	Positive
Nitrite	Neg / Negative	Confirmed negative
Leukocytes	Small 70, Moderate 125, Large 500	Positive (+ or greater)
Leukocytes	Trace 15	No source-matched update; not positive and not negative
Leukocytes	Neg / Negative	Confirmed negative
Blood	Hemolyzed 10, Small 25, Moderate 80, Large 200	Positive (hemolyzed trace or greater)
Blood	Non-hemolyzed 10	No source-matched update; not used as the Little threshold
Blood	Neg / Negative	Confirmed negative
Any	Unavailable, invalid, or unreliable	LR 1.00; no update

Strip profile used in software: URS-10T. The study threshold mapping above is frozen in the implementation. The physical product's exact manufacturer, lot, IFU revision, and reaction-time instructions must still be recorded for each validation run; they must not be inferred from the generic URS-10T name.

4. Evidence Library

These values are retained to show the cited evidence. Not every value is used in the implemented equation.

Finding	LR+ / present	LR- / absent	Population and reference standard	Implemented?
Nitrite	5.50	0.56	Symptomatic women; review table based on culture-referenced evidence	LR+ only through Group A hierarchy
Leukocyte esterase	1.40	0.40	Symptomatic women; review table based on culture-referenced evidence	LR+ only through Group A hierarchy
Dipstick blood	1.70	0.89	Symptomatic women; review table based on culture-referenced evidence	LR+ only through Group A hierarchy
Dysuria	1.30	0.67	Symptomatic women; culture-referenced evidence review	Present LR only
Urgency	1.20	0.75	Symptomatic women; culture-referenced evidence review	Present LR only
Frequency	1.10	0.71	Symptomatic women; culture-referenced evidence review	Present LR only
Dysuria with urgency	1.50	0.44	Symptomatic women; culture-referenced evidence review	Present LR only
Visible hematuria	2.00 (95% CI 1.3-2.9)	Not assigned	Meta-analysis of signs/symptoms in women with acute uncomplicated UTI	No; context only
Vaginal discharge	0.30 (95% CI 0.1-0.9)	Not assigned	Same Bent review	No; alternate-cause route
Vaginal irritation	0.20 (95% CI 0.1-0.9)	Not assigned	Same Bent review	No; alternate-cause route
Nitrite plus leukocytes or blood	7.20	Not assigned	427 women with suspected UTI; laboratory diagnosis using European urinalysis guideline	Yes
Nitrite, leukocytes, and blood all negative	Not assigned	0.22	Same Little validation study	Yes

The AAFP table's displayed post-test probabilities assume a 50% pretest probability. Bent et al. reports an approximate 50% probability among women presenting with one or more UTI symptoms. This supports 50% only as a provisional starting assumption for the gated study-like population.

5. Exact Implemented Scoring Hierarchy

5.1 Group A - Dipstick Pattern

Apply the first matching rule only.

Rule	Pattern	Candidate LR
A1	Nitrite positive plus source-threshold leukocytes or blood positive	7.20
A2	Nitrite positive without either additional source-threshold positive finding	5.50
A3	Nitrite not positive or unavailable; source-threshold blood positive	1.70
A4	Nitrite and blood not positive; source-threshold leukocytes positive	1.40
A5	Nitrite, leukocytes, and blood all confirmed negative	0.22
A6	No rule above matches	1.00; no update

5.2 Group B - Urinary Symptom Pattern

Apply the first matching rule only.

Rule	Pattern	Candidate LR
B1	Dysuria and urgency present	1.50
B2	Dysuria present	1.30
B3	Urgency present	1.20
B4	Frequency present	1.10
B5	None of the weighted symptoms is reported	1.00; no update

Visible hematuria and suprapubic pain can satisfy the symptomatic eligibility gate but have no numerical factor in v1.1. Visible hematuria triggers a separate consultation-context message.

5.3 State Handling

Input state	Numerical treatment
Present	Apply the approved present LR if the variable is in Groups A or B.
Confirmed absent	No symptom-negative LR is used in v1.1. Exact dipstick negatives may satisfy A5.
Not selected / not asked / unknown	LR 1.00; no update.
Unavailable / unreliable	LR 1.00; no update.
Not applicable or outside population	Do not calculate.

6. Equation

```
prior probability = 0.50
prior odds = 0.50 / (1 - 0.50) = 1.00
```

```
posterior odds = prior odds x selected Group A LR x selected Group B LR
posterior probability = posterior odds / (1 + posterior odds)
```

Example for an eligible patient with nitrite plus leukocytes positive and both dysuria and urgency:

```
posterior odds = 1.00 x 7.20 x 1.50 = 10.80
research posterior = 10.80 / 11.80 = 0.915 or 91.5%
```

This arithmetic is correct, but the combined equation is a URITECT design assumption. The cited sources do not validate the exact multiplication of Group A and Group B as a single calibrated clinical model. Dependency, transportability, and calibration require culture-confirmed patient-level validation and statistical review.

7. Separate Action Pathways

Trigger	App action
Eligibility incomplete or outside intended population	Consultation with a healthcare professional is suggested.
Vaginal discharge or irritation	Other conditions may cause similar symptoms. Consultation is suggested; no ordinary posterior.
Visible hematuria	Consultation is suggested; no hematuria LR is added.
Fever with flank pain	Prompt medical consultation is suggested.
Nausea/vomiting with fever or flank pain	Prompt medical consultation is suggested.
Vomiting prevents hydration or oral medication	Prompt medical consultation is suggested.
Confusion, fainting, or severe weakness with a urinary finding	Prompt medical consultation is suggested.
Nausea/vomiting alone	Consultation is suggested; it does not independently activate the prompt pathway.
No usable source-matched evidence	Insufficient evidence. The 50% prior is not shown as a patient result.

The UTI pathway is separate from the renal follow-up engine. Neither pathway recommends antibiotics or diagnoses disease.

8. Display and Audit Rules

- No 20%/80% thresholds exist in v1.1.
- No Lower, Intermediate, or Higher UTI probability label is produced.
- An eligible calculation displays the provisional percentage, 50% prior, and each applied LR.
- The display label is: Bayesian UTI screening estimate - screening support only, not a diagnosis or treatment recommendation.
- The calculated posterior and factors are also stored in the local scan audit record.
- The model version and calculation status are stored with the result.
- A blocked or insufficient case stores no patient-specific posterior.

9. Validation Boundaries

Two registered medical technologists may assess relevance, clarity, manufacturer-label mapping, workflow feasibility, missing-result handling, and laboratory limitations. Content Validity Index agreement does not establish diagnostic accuracy or probability calibration.

Before any clinical probability claim, the complete model requires:

- Physician review of population gates, symptom meaning, actions, and safety wording.
- Statistician or clinical prediction-model review of priors, dependence, calibration, and reporting.
- Prospective culture-confirmed patient-level validation, including calibration assessment and confidence intervals.
- Validation of the exact supported strip product, IFU revision, reading times, phone/camera workflow, and classifier errors.

10. Medical Technologist Review Checklist

Rate each item: 4 = highly relevant/clear, 3 = relevant/clear with minor revision, 2 = major revision needed, 1 = not relevant/unclear.

ID	Review statement	1	2	3	4	Comment
C1	The intended population gate is understandable.	☐	☐	☐	☐	
C2	Unchecked eligibility items are clearly treated as unknown.	☐	☐	☐	☐	
C3	Nitrite positive mapping is appropriate for the supported strip.	☐	☐	☐	☐	
C4	Leukocyte + threshold mapping is clear.	☐	☐	☐	☐	
C5	Trace 15 leukocytes are correctly excluded from source-threshold positivity.	☐	☐	☐	☐	
C6	Hemolyzed-trace blood mapping is clear.	☐	☐	☐	☐	
C7	Non-hemolyzed trace blood is appropriately excluded from the Little threshold.	☐	☐	☐	☐	
C8	Missing, unreliable, and negative results are clearly distinguished.	☐	☐	☐	☐	
C9	One mutually exclusive dipstick factor reduces duplicate counting.	☐	☐	☐	☐	
C10	Visible hematuria is appropriately retained as context without a numerical LR.	☐	☐	☐	☐	
C11	Vaginal symptoms are appropriately routed to consultation.	☐	☐	☐	☐	
C12	The systemic-symptom action wording is clear and safe.	☐	☐	☐	☐	
C13	The 50% prior limitation is stated clearly.	☐	☐	☐	☐	
C14	The lack of local calibration is stated clearly.	☐	☐	☐	☐	
C15	The distinction between CVI and clinical validation is clear.	☐	☐	☐	☐	
C16	The workflow is feasible for intended rural/offline use.	☐	☐	☐	☐	

Overall decision: ☐ Accept as written ☐ Accept with minor revisions ☐ Major revision required ☐ Reject

Reviewer name and credentials: _____ Registration/license number: _____
 _____ Institution: _____ Signature: _____
 _____ Date: _____

11. Required Product and Validation Record

Field	Entry
Strip manufacturer	_____
Exact product/catalog number	_____
Lot number and expiry	_____
IFU revision/date	_____
Nitrite reaction time	_____
Leukocyte reaction time	_____
Blood reaction time	_____
Phone model / Android version	_____
App version / model version	1.3.0+4 / uti_bayesian_lr_v1_1_20260926

12. References

1. Kurotschka PK, Gagyor I, Ebell MH. Acute Uncomplicated UTIs in Adults: Rapid Evidence Review. American Family Physician. 2024;109(2):167-174. <https://www.aafp.org/afp/2024/0200/acute-uncomplicated-utis-adults>
2. Little P, Turner S, Rumsby K, et al. Developing clinical rules to predict urinary tract infection in primary care settings. British Journal of General Practice. 2006;56(529):606-612. <https://pmc.ncbi.nlm.nih.gov/articles/PMC1874525/>
3. Bent S, Nallamotheu BK, Simel DL, Fihn SD, Saint S. Does this woman have an acute uncomplicated urinary tract infection? JAMA. 2002;287(20):2701-2710. <https://pubmed.ncbi.nlm.nih.gov/12020306/>

13. Final Reviewer Acknowledgment

I understand that this review evaluates content and implementation clarity. It does not establish that the combined posterior is calibrated or clinically valid, and it does not replace physician, statistical, or culture-confirmed validation.

Reviewer signature: _____ Date: _____